

PRODUCT STRATEGY FOR THE COMPUTATIONAL INJURY ANALYSIS GROUP

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ABSTRACT. // DRAFT - WORK-IN-PROGRESS - NOT FOR RELEASE // As of this writing, the intended audience for this brief is limited to Douglas Dederman and the author.

1. PURPOSE

The purpose of this brief is to articulate a product strategy for the Computational Injury Analysis Group within Terminal Ballistic Department (Department 5421) at Sandia National Laboratories.

This brief is in response to Wallace Bow,¹ which stated that the Center (Integrated Military Systems, Center 5400) solicits information contained in this brief from its organizational reports so that the Center can create a list, called STIPL, Science and Technology Integrated Priority List, which then allows the Center “to define and prioritize our most pressing technological issues to guarantee that our innovation investments are being properly allocated.” The end goal of effort and funding allocation is “to ensure that our products remain superior.”

2. BACKGROUND

To effectively articulate a product strategy for the Computational Injury Analysis Group, some background is helpful.

2.1. **Who.** The group is led by Douglas A. Dederman (Manager of Department 5421). Paul A. Taylor is the senior member. Candice F. Cooper, Shivonne Haniff, and Chad B. Hovey are members.

2.2. **What.** The group has product offerings in modeling and simulation, and scientific publications. The group’s model is a high-fidelity head, neck, and torso human body model.

2.3. **When.** There have been two main Laboratory Directed Research and Development (LDRD) grants, the first in FY2007 for one year and the second in FY2013 for three years.

¹See email 2016-10-31 1433 MST.

2.3.1. *Traumatic Brain Injury.*

- FY2007 marked an investment of \$99,000 sponsored by Defense Systems and Assessments (Division 5000), for Paul A. Taylor, Principal Investigator, John B. Aidun, Product Manager, to investigate a connection between blast and traumatic brain injury (07-1614 Proposal Summary, LDRD Project 113675 “Traumatic Brain Injury”). The \$99,000 funding amount was recorded as 0.30 Full Time Equivalent (FTE) in Center 1400, with zero funding requests for out years FY2008 and FY2009.

2.3.2. *Wound Ballistics Modeling for Blast Loading, Blunt Force Impact, and Projectile Penetration.*

- FY2013 marked a funding request of \$380,000 for labor (Taylor at 0.5 FTE, Ludwigsen at 0.1 FTE, Brundage at 0.1 FTE) and \$35,000 for acquisition of medical image data and hiring part-time students to segment the medical images, total \$415,000.
- FY2014 requested \$392,000 for labor at the same FTE allocation as in FY2013 (\$12,000 or 31.6 percent YoY increase) and \$4,000 for miscellaneous supplies, total \$396,000.
- FY2015 requested \$396,000 for labor at the same FTE allocation as in FY2014 (\$4,000 or 1.02 percent YoY increase), \$4,000 for miscellaneous supplies, and \$4,000 for travel to present at scientific conferences, total \$404,000.
- Total for second LDRD investment: \$1,215,000.
- Total for both LDRD investments: \$1,314,000.

See Taylor IDAPR LDRD Quad New_Initial.v7.pptx and LDRD Project 165554.

2.4. Other investments. ONR funding (Bentley, amount, time) to be determined. Other funding to be determined.

3. CONTEXT

In this section, we answer the questions, “What is the group’s strategy and how does it integrate with the larger mission and vision of Sandia National Laboratories.” To answer these questions, we first establish the mission and vision of the Laboratory, Division, Center, and L2/L1 organizations. Then, we examine how the group’s purpose integrates into these larger organizations.

3.1. Sandia National Laboratories. Sandia’s mission states, “The synergy and interdependence between our nuclear deterrence mission and broader security missions forge a robust capability base and empower us to solve complex national security problems.”²

Sandia’s vision states, “On behalf of our nation, we anticipate and solve the most challenging problems that threaten security in the 21st century.”

²See <http://strategicplan.sandia.gov/vision-mission-values.html>.

3.2. Defense Systems and Analysis. Within Sandia, the Defense Systems and Assessments (DSA, Division 5000) states its mission as “We shape the future of national security by delivering tools, technologies, and pathfinder solutions that exploit the vulnerabilities of U.S. adversaries. By leveraging the Lab’s research and development in advanced science, technology, and systems engineering we create and advance capabilities that are synergistic with our nuclear weapons mission.”³

DSA’s vision states, “We transform research into game changing solutions that provide the warfighter and intelligence community with unmatched superiority.”

3.3. Integrated Military Systems. Within DSA, Integrated Military Systems (IMS, Center 5400) states its mission as “Integrated Military Systems develops and demonstrates game-changing systems and technologies to solve our warfighters’ most challenging, urgent, and high-risk problems—to defend our nation and defeat threats to national security.”⁴

IMS states its vision as “Render exceptional service in the national interest and seize opportunities that protect our warfighter and promote national security.”

3.4. Strike and Aerospace Systems. Within IMS, the Strike and Aerospace Systems (SAS, Organization 5420) states its mission as “Support the DoD and other Federal agencies at Sandia’s NWSBU Advanced Systems and Technology Program with technologies focused on high-speed reentry, precision, navigation, earth-penetrating weapons, conventional munitions, and guided flight systems.”⁵

As of this writing, I have not found a vision statement.

3.5. Terminal Ballistics Technology. Terminal Ballistics Technology (TBT, Organization 5421) states its mission as “Design, develop, and demonstrate terminal ballistics enabling technologies that enhance survivability and functionality of lethal and nonlethal projectiles/probes for characterization, defeat or protection of military and nonmilitary targets.”⁶

TBT has a programmatic relationship with Military Systems and Technologies (MST, Organization 5440). MST states [with a style of a mission statement], “The people of 5440 apply science-based systems engineering, integrated technologies, and advanced analytics to enhance military operations and mission effectiveness.”⁷

4. MISSION AND VISION

4.1. In a nutshell.

³See <https://sharepoint.sandia.gov/sites/dsa-div-5000/Pages/default.aspx>.

⁴See https://sharepoint.sandia.gov/sites/IntegratedMilitarySystems/About_Us.aspx.

⁵See http://info.sandia.gov/imsdc/dept15420/5420_home.htm.

⁶See <http://info.sandia.gov/imsdc/dept5431/index.htm>.

⁷See http://info.sandia.gov/imsdc/dept5440/5440_home.htm.

- The Laboratory’s mission focuses on the solution of complex national security problems with nuclear deterrence and the derivatives therefrom, with the vision of solving these problems into the 21st century.
- The DSA mission focuses on national security solutions by leverage of research and development that is synergistic to the nuclear deterrent. The vision articulates warfighter and intelligence community superiority from DSA’s transformation of research into solutions.
- The IMS mission focuses on national security through solution of warfighters’ problems and visualizes that the result will be game-changing.
- The SAS mission focuses on the launch, fly, navigate, and penetrate of high speed weapon systems.
- The TBT mission focuses on survivability and functionality of ballistics technology.
- The MST mission focuses on enhanced military operations and mission effectiveness.

4.2. Connection to the customer. The national security mission is supported at all levels of the organization. The execution of that mission can be broadly subdivided into military and intelligence customers. Service of the military customer can be further divided into area of application: air, land, and sea (e.g., Air Force, Army, and Navy). Cross-cutting these application environments, service to national security can be categorized into human-exposed and human-non-exposed applications. With the human-exposed category, the Computational Injury Analysis Group focuses on the two most important exposure areas: blast and ballistics.

4.3. Computational Injury Analysis Group mission and vision. With this context of the organizational missions and our connection to the organization’s structure, we now state our mission and vision.

Mission: We analyze injury from blast and ballistics to help the nation protect the U.S. warfighter.

Vision: To significantly reduce the risk of death or serious injury from blast and ballistics by 2027.

4.4. Exploration of word choice. To better understand the mission and vision statements, let us now weigh out each word and explore not only the denotation, but also the connotation.

Why computational? Because the core competency of our group is within modeling and simulation. We, by design, exclude experimental work from efforts we undertake. Our interaction with experimental groups, therefore, is collaborative but not organizational.

Why injury? Because injury, or specifically, the reduction or elimination of injury, is fundamental to the nation’s ability to marshal national security resources that require a human in-the-loop. We emphasize the word “injury” because the analysis goes beyond demonstration of engineering response (e.g., stress, strain) of human tissue and human personal protection (e.g., bullet proof vest) to war environments. The analysis connects engineering response to human threshold of injury to provide evidence-based prediction on injury severity: minor, moderate, serious, severe (survivability likely) critical (survivability unknown), and maximum (not survivable).⁸

Why analysis? Because our main product is the advancement of science to understand injury mechanisms. There is an implicit priority of analysis first, development second. Specifically, our primary focus will be to use *existing* simulation tools (e.g., ALEGRA, CTH, Sierra) and *existing* models (e.g., BOB, Global Human Body Models Consortium,⁹ and Total Human Model for Safety¹⁰) to provide insight and answers to scientific problems and questions. Fundamentally, our focus with tools and models will be as a means to an end, and not as the end itself.

Why blast and ballistics? Because the exposure of the warfighter to bombs and high velocity munitions continue to be a specific and relevant risk to the U.S. warfighter. We emphasize that blast and ballistics is distinct from general impact because the former has characteristically large forces and velocities that occur in extremely small times. This confluence of scales creates an environment that is extremely hostile to human viability.

Why by 2027? Because we believe pursuit of significant reduction to the risk of death or serious injury to the U.S. warfighter will remain an enduring research endeavor over the next decade, worthy of considerable investments of time and resources. It is interesting to conceive of autonomous systems (e.g., robots) taking the place of a human soldier on the war field. In this scenario, our mission may evolve, but our vision would remain unchanged. Our mission, therefore, might shift to protection of the human surrogate from malfunction or obliteration secondary to exposure to blast or ballistics. In this way, we have attempted to be forward-thinking in our vision, while calibrating a mission to important, present-day needs.

5. STRATEGY

5.1. Motivation. In a global context, we direct our efforts to protect people who promote peace. The specific implementation of this ideal is through the U.S. warfighter, called upon

⁸See the Abbreviated Injury Scale (AIS), by the Association for the Advancement of Automotive Medicine (AAAM).

⁹See <https://youtu.be/u8l9Lk1jCU>

¹⁰See www.lstc.com/thums

by service to country to fight tyranny and the abuse of power. “Life, liberty and the pursuit of all who threaten it,” the U.S. Navy’s slogan, adroitly encapsulates the warfighter’s mission. To that end, it is incumbent on those who benefit from a warfighter’s risk taking that we help to minimize exposure to that risk. Specifically, as long as blast and ballistics remain an inherent element of conflict, we should equip the warfighter with the best available protection from the same.

Fundamental to the best available protection paradigm is the understanding of how blast and ballistics imperil the warfighter’s survivability by exposing the warfighter to traumatic brain injuries that are either focal (penetrating through the warfighter’s helmet or face, impact to a side of the helmet or face) or diffuse (causing localized shear damage to the neural microstructure of the brain).

5.2. Background (continued from Section 2). Our strategy is build from our extraordinary success since our first funding in FY2007. However, before we chart the path forward, it is important to lay out, in an abbreviated form, the foundation from which we build and articulate how our group is unique in the world and thus positioned to offer the nation a significant value proposition.

What is our current product?

- We have developed and validated human tissue constitutive models for use in Sandia’s Sierra and CTH computational code bases. *[Need to be specific as to what models, and reconcile previous emphasis on tool use, not tool building (analyst role versus developer role).]*
- We have developed and validated a human anatomical model of the head, neck, and torso, having 1 millimeter resolution, based on the Visible Human Project. *[Again, need for reconciliation of strategy: tool use versus tool development.]*
- We have postulated to the scientific community, through peer-reviewed journal publications and scholarly conferences, that critical thresholds of deviatoric strain energy in the human brain secondary to blast correlate with traumatic brain injury (TBI) and clinical functional magnetic resonance imaging (fMRI) data. Moreover, we have suggested that the brain has a compensatory mechanism ability, where a non-injured part of the brain will attempt an uptake in function to compensate for the damaged areas of the brain. This compensatory strategy is well-known in human ambulation and gait mechanics, but remains relatively nascent in brain science.
- Finally, we have demonstrated that the presence of air bubbles in cerebral spinal fluid (CSF) after blast to the brain result in cavitation, which is a damage mechanism to the myelin sheath, the protective covering of nerve fibers.
- *Include Candice’s viscous criteria effort once Doug and Paul have cleared Chad to know details about it.*

Why have we focused on the head, neck, and torso? Because our strategy focuses our limited resources on serious-to-fatal injuries. While fatal injuries can still occur secondary to injury to the upper or lower extremities, the prevalence in these body regions is relatively low compared to injuries to the head, neck, and torso.

6. COMPETITIVE LANDSCAPE

How are we uniquely positioned? Competitive analysis - to be completed.

7. PLAN

There will be four main thrusts to the FY2017 plan, corresponding to each of the four members of the group led by Dederman.

7.1. Continuity. Paul Taylor will lead a continuity effort, motivated by his anticipated retirement and the group's desire to minimize the "brain drain" caused by his departure. Specific deliverables are

- (1) Complete funded research obligations, including presentations and scholarly journal articles.
- (2) Complete and convey documentation of human tissue model formulation and implementation.
- (3) Mentor Cooper/Haniff/Hovey with the goal of making them self-sufficient members of the research team:
 - (a) Establish professional connections from Taylor to balance of group.
 - (b) Coach proficiency with tool use (e.g., CTH) in all members.

7.2. Torso Mechanics. Candice Cooper will lead efforts to

- (1) Verify and validate human torso model with known cadaver impact studies. Demonstrate the human torso model can reproduce known viscous criteria benchmarks.
- (2) Work with a personal protection customer to solve one significant torso injury problem caused by blast or ballistics.
- (3) Publish in scholarly journals
 - (a) The validation of the torso model.
 - (b) The solution to an industry problem.

7.3. Brain Micro-Mechanics. Shivonne Haniff will lead efforts to

- (1) Compare and contrast three micro-mechanical brain injury mechanisms:
 - (a) Neuron damage from cavitation

- (b) Neuron damage macro-level head rotations in sagittal, frontal, and transverse plane motions.¹¹
- (2) Harmonize macro and phenomenological brain injury models (e.g., G and T) with micro-mechanical brain injury models.

7.4. Head-Neck Interface, Group Development, Business Development. Chad Hovey will lead efforts to

- (1) Atlanto-occipital dislocation (AOD)
 - (a) Large-slip contact of occipital-C1 junction
 - (b) Support for gross head motion for Haniff's work
 - (c) Pilot ejection head and neck injury mitigation
- (2) Develop group's identity and purpose of computational injury analysis from blast and ballistics.
- (3) Develop economies of scale within group, so that research efforts multiply, not just add.
- (4) With Doug Dederman, develop a sustaining customer base.

7.5. STIPL proposed paragraph. We will strengthen our contribution to warfighter protection from blast and ballistics by leveraging our existing modeling and simulation investments to pursue deeper understanding of three scenarios that threaten warfighter survivability: (1) traumatic brain injury from blast that causes microscale neuron damage from a cavitation mechanism; (2) traumatic brain injury from ballistic aircraft ejection that causes diffuse axonal injury from a high rotation rate and acceleration mechanism; and (3) traumatic aortic disruption from behind armor blast and ballistic impact. In patients with multi-trauma, head injury and aortic disruption are the leading and second most common risk of death. Thus, in our vision to significantly reduce the risk of death and serious injury to the warfighter from blast and ballistics by 2027, we position ourselves to produce maximum scientific impact by addressing the two leading causes of traumatic fatality.

8. PMP PLANNING

Regarding the requests to

- identify three to five objectives covering work execution, business development, and model behavior, and
- identify the metric used to determine to what degree the objectives are met,

for FY2017, I propose the following: project, funding, and team development.

¹¹The sagittal plane rotation is the "yes" rotation, with chin up-and-down movement. The frontal plane is the "shoulder-to-shoulder" rotation, with alternating ear-to-shoulder movements. The transverse plane rotation is the "no" rotation, with chin alternating left-to-right movements.

8.1. **Project Development.**

- Objective 1(a) Identify a customer (e.g., USAF 711th Human Performance Wing) with a problem relevant to the customer (head and neck injury from pilot ejection) and within the mission scope of the Computational Biomechanics Analysis Group (injury from “blast” either from ejection seat or subsequent air blast and “ballistics” as the pilot, post ejection, is now a ballistic object).
- Objective 1(b): Create or use an existing model in a simulation code (e.g., Sandia or otherwise) to predict head rotation metrics (angular velocity and angular acceleration) and correlate these metrics to known human thresholds for Diffuse Axonal Injury (DAI).
- Objective 1(c): Use these metrics as inputs to Haniff’s continued work on microscale traumatic brain injury (TBI).

Metrics used to determine to what degree the objectives are met:

- Get endorsements from Dederman, Taylor, Cooper, Haniff, Helmich, and Roesler that the strategic direction for the group is specific, measurable, relevant, achievable, and time-driven.
- Get agreement from client that the problem remains unsolved, and that the proposed approach, if executed, would solve the client’s problem and provide a significant value proposition.
- Produce simulations that are validated with experiments that demonstrated onset of DAI.
- Have simulation outputs be successfully integrated into a microscale model for TBI.

8.2. **Funding Development.** Next,

- Objective 2: Bring in new funding for group through LDRD and other granting agencies.

Metrics used to determine to what degree the objectives are met:

- If the funding proposals are in alignment with Center 5400 mission, vision, and strategies.
- The amount of funding brought (absolute dollars).
- The number of FTEs supported.
- The percentage of funding brought in relative to competitors within the same market space (relative dollars).

8.3. **Team Development.** Finally

- Objective 3: Develop the group’s identity, purpose, and scientific impact, making sure economies of scale are achieved (sum of effort exceeds individual contributions of each member).

Metrics used to determine to what degree the objectives are met:

- Recognition of the group within each group member (self-identity)

- Recognition of the group within the Department
- Recognition of the group within the Center, Division, and Laboratory
- Recognition outside of Sandia (e.g., other laboratories, journal publications, news articles)
- Funding above and beyond Objective 2 funding, secondary to recognition efforts.

9. RISKS OF PLAN

To be completed.

- Sufficient diversity (avoid all eggs in one basket, coverage of ALEGRA, CTH, Sierra) versus economies of scale (all member speak three different languages, so communication and collaboration is inhibited). This is the classic “build horizontal” versus “build vertical” trade off.